BREWSTER SMYERS

Marblehead, MA | 781-513-7141 | brewstersmyers@outlook.com
linkedin.com/in/brewstersmyers

EDUCATION

Bucknell University | B.S. Mechanical Engineering, expected May 2028

Minors: Mathematics and Business Analytics

Relevant coursework: Dynamics, Fluid Dynamics, Thermodynamics, Manufacturing Processes, Foundations in Aircraft Design, Differential Equations, Probability and Statistics

TECHNICAL SKILLS

SOLIDWORKS (CSWA) • FEA simulation • Mastercam • MATLAB • Python • Java • Carpentry • Microsoft Office

ENGINEERING PROJECTS

Outdoor Landmark Chair | Independent design and fabrication

- Designed an oversized braced-frame outdoor chair in SOLIDWORKS with a preliminary stress study; built it alone in two days on an approximately $1,000 budget. Seats four to five people and remains in use.

Nantucket Coaster Injection Mold | Manufacturing Processes, Fall 2026

- Designed a coaster and its aluminum injection mold in SOLIDWORKS; wrote the process plan and Mastercam toolpaths (5 tools, 880 lines of G-code). The mold was machined on a Haas mill from my G-code.

Bronze Cast Rook | Manufacturing Processes, Fall 2026

- Designed and 3D printed a pattern, built the wax gating, and cast it in bronze by investment casting; measured the result against CAD (width 3.36% smaller, mass about 14% below prediction).

90° Pipe Brace Bracket | Independent FEA study, November 2026

- Ran a static FEA of a welded steel pipe bracket at 1,000 N in SOLIDWORKS Simulation; confirmed the 140 MPa bend stress by hand (factor of safety about 1.6) and traced a 376 MPa peak to a sharp-corner artifact.

Underwater Vehicle and Water Tunnel | Team lead, Fluid Dynamics project, Fall 2026 (in progress)

- Lead a six-person team designing an unmanned underwater vehicle hull and a miniature water tunnel to test a scale model; own the tunnel design and completed hull hydrostatics and buoyancy calculations by hand.

Aerobatic Biplane | SOLIDWORKS and MATLAB concept study, Fall 2026

- Modeled a two-seat biplane in SOLIDWORKS; adapted a MATLAB analysis of lift, drag, fuel use, power, and trim.

LEADERSHIP AND SERVICE

Infrastructure and Social Chair | Sigma Phi Epsilon, Bucknell University

- Designed and constructed modular risers, seating, and stages; planned events for 100+ attendees, coordinating logistics, vendors, and member participation.

Teaching and Mentoring Assistant | LEAP for Education, Summers 2022 and 2023

- Supported students through seasonal educational outreach and mentoring.

WORK EXPERIENCE

Bartender | The Nantucket Hotel & Resort, Nantucket, MA | 2026

- Delivered high-volume service across the bar, pool, guest rooms, and restaurant; prepared and organized service stations for weddings and special events.

Bartender | The Landing, Marblehead, MA | 2022–2025

- Managed service for 13 bar seats and three high-top tables while coordinating food and drink orders, preparation, and restaurant-wide support.

ACTIVITIES

Club Ultimate Frisbee Founder • Skiing, Cycling, Hiking • Completed an indoor 70.3-mile triathlon and raised $500+ for men’s mental health awareness • Avid New England Patriots Fan